SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	MURSHIDHA.M	Reg. No.:	192571007
Section / Batch:	A	Date:	20/07/2026

Code of Conduct

I MURSHIDHA.M *(Name / Reg No)* *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Murshidha

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
	GRAND TOTAL:	100 Marks	

16	192521194	LOKESH WARAN T	1. Consider a linked list representing the sales of different products in a store for a week. Each node of the linked list contains the product ID and the number of units sold on a specific day. Implement a linked list data structure and provide functions to: Insert new sales data for a given product on a specific day. Calculate the total sales for each product at the end of the week. Find the product with the highest sales and its corresponding day. Display the sales data for a specific product for the entire week. Use your own data for product. 2. A calculator application receives the following expression: (8+2)*(6-4)/2
17	192372296	MANGADHODDI DEEPIKA	1. A hospital treats patients on a First-Come, First-Served (FCFS) basis. Patients arrive in the following order: P1 (General) P2 (Emergency) P3 (General) P4 (General) P5 (Emergency) *Questions*: ➢ If a normal queue is used, what will be the treatment order? ➢ Explain why this approach is not suitable. ➢ Suggest a better queue-based solution and justify your answer 2. A calculator application receives the following expression: (10+5) * (3−4)/2
18	192513069	MADHUMITHA K	1. Find the difference between the minimum and maximum values in a single linked list having 34->77->22->55->66->11-> in the list. 2. The double linked list has two links in each node as “prev”, “data” and “next”. Display the elements from DLL in forward and reverse order and find the average of those ‘n’ elements.
19	192525318	MADINENI DHANUSH KUMAR	1. Print all the odd and even counts present in the single linked list having 10 elements in the list 77->88->22->11->66->27->55->65->18->10-> 2. Search linearly a key elements K=20 present in the list of single lined list. If so print “YES” else print “NO”.
20	192571007	MURSHIDHA M	1. A supermarket has only one billing counter. Customers arrive in the order: C1, C2, C3, C4, C5. After C2 is served, a senior citizen arrives. *Questions:* ➢ Explain how the queue changes if a normal queue is used. ➢ Design a queue-based solution that prioritizes senior citizens while maintaining fairness. 2. Assume a single character is present in the data field of a Single Linked List with 6 nodes as S- > I -> M -> A -> T -> S. Write a

38	192524251	VAISHNAV M	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. ➢ Analyze why this occurs. ➢ Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	1. Consider the singly linked list: 10 → 20 → 30 → 40 → 50 Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: 1 → 3 → 5 → 7 L2: 2 → 4 → 6 → 8

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: ________	Signature: ________	Signature: ________
Date: ________	Date: ________	Date: ________

CSA0305- DATA STRUCTURES

MURSHIDHA.M   ANALYTICAL PROBLEM   192571007

1. A Supermarket has only one billing counter. Customers arrive in order: C1, C2, C3, C4, C5. After C2 is served, a senior citizen arrives.

Questions:

→ Explain how the Queue changes if a normal Queue is used.

→ Design a Queue-based solution that prioritizes senior citizens while maintaining fairness.

Problem Statement

A Supermarket has only one billing counter customers arrive in following order:

C1, C2, C3, C4, C5

After C2 is served, a senior citizen (SC) arrives.

(a) Queue changes using Normal Queue.

A normal Queue follows the fifo (first in first out) principle

Initial Queue

Front → C1 → C2 → C3 → C4 → C5 → Rear

Step 1: C1 is served

Front → C2 → C3 → C4 → C5 → Rear

Step 2: C2 is served

Front → C3 → C4 → C5 → Rear

Step 3: Senior citizen arrives.

Since a normal Queue does not give priority, the senior citizen joins at rear.

Front → C3 → C4 → C5 → SC → Rear

Final Serving order

C1 → C2 → C3 → C4 → C5 → SC

CONCLUSION

IN a normal Queue, the Senior citizen has to wait unit all customers who arrived earlier have been Served.

SERVING ORDER:

C1 ⟶ C2 ⟶ SC ⟶ C3 ⟶ C4 ⟶ C5

2. Assume a single character is present in the data field of a single linked list with 6 nodes as S -> I -> M -> A -> T -> S. Write function to print the same in reverse order using stack data structure.

```
# include <stdio.h>
# define MAX 6
Struct Node
{
  char data;
  Struct Node *Next;
};
Char stack [MAX];
int top = -1;
void push(char ch)
{
  stack [++ top] = ch;
}
Char pop ()
{
  return stack [top--];
}
```

(b) Queue-based solution with priority for Senior citizens.

Maintain two Queues:

- Senior Queue (SQ) - Stores senior citizens
- Normal Queue (NQ) - Stores other customers

Algorithm

1. if the Arriving customer is a Senior citizen, insert into SQ
2. Otherwise, insert into NQ
3. During service:
   - if SQ is not empty, serve from SQ
   - Otherwise, serve from NQ.

WORKING:

initially

SQ : Empty

NQ : C1 ⟶ C2 ⟶ C3 ⟶ C4 ⟶ C5

After serving C1 and C2

SQ : Empty

NQ : C3 ⟶ C4 ⟶ C5

SENIOR CITIZENS ARRIVES

SQ : SC

NQ : C3 ⟶ C4 ⟶ C5.

```
void Print Reverse (Struct Node *head)
{
 Struct Node *temp = head;
While (temp != NULL)
{
  Push (temp -> data);
  temp = temp -> Next;
}
while (top != -1)
{
 printf("%c", pop());
}
}
int main ()
{
 Struct Node n1, n2, n3, n4, n5, n6;
n1.data = 'S';
n2.data = 'I';
n3.data = 'M';
n4.data = 'A';
n5.data = 'T';
n6.data = 'S';
```

```
n1.next = &n2;
n2.next = &n3;
n3.next = &n4;
n4.next = &n5;
n5.next = &n6;
n6.next = &null;
PrintReverse (&n1);
return 0;
}
```

Linked List

S → I → M → A → T → S

After pushing:

Top → S
I
M
A
T
S

popping gives:

S → T → A → M → I → S